Phase-Associative Memory: Sequence Modeling in Complex Hilbert Space

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The separability assumption that underlies both classical physics and the design of modern language models has been challenged by experiments demonstrating that semantic interpretation in humans and large language models produces correlations that violate Bell-type inequalities. If the correlational structure of language is non-classical, the natural computational formalism for modeling it is a complex Hilbert space in which representations carry phase, similarities are computed through the conjugate inner product, and interference is an intrinsic property of the algebra. We present Phase-Associative Memory (PAM), a recurrent sequence model that operates entirely in this space: associations accumulate in a complex matrix state $S_t \in \mathbb{C}^{d\times d}$ via outer products, and retrieval uses the conjugate inner product $K_t^* \cdot Q_t/\sqrt{d}$, whose selectivity depends on phase alignment between stored and queried representations. At ∼100M parameters on WikiText-103, PAM reaches validation perplexity **30.0**, within ∼10% of a matched transformer (**27.1**) trained under identical conditions, despite 4× arithmetic overhead and no custom kernels. We trace the experimental path from vector-state models, where holographic binding fails due to the $O(1/\sqrt{n})$ capacity degradation of superposed associations, to the matrix state that resolves it, and discuss what the competitiveness of a fully complex-valued architecture implies for the choice of formalism in the study and modeling of natural language.

INTRODUCTION

The assumption that the constituents of a system can be analyzed independently of one another and of the conditions under which they are observed has served as a foundational premise of empirical science since the seventeenth century [1, 2]. The intellectual ground for this assumption was prepared over several centuries by the incorporation of Aristotelian metaphysics into Christian scholastic philosophy, most influentially by Thomas Aquinas in the thirteenth century [3], whose synthesis treated the natural world as composed of distinct substances possessing essential properties that could be enumerated and studied in isolation from the observer [4]. Galileo Galilei [5], whose mathematical treatment of motion in the early seventeenth century demonstrated that physical law could be expressed as quantitative relationships between measurable properties of bodies, stripped away the Aristotelian concern with purposes and essences in favor of a description in which the state of an object could be fully specified by quantities that exist independently of the person measuring them. Francis Bacon, writing contemporaneously, formalized the principle that nature is best understood through controlled intervention, arguing in the *Novum Organum* [6] that the isolation of individual variables through systematic experimentation was the proper method for uncovering causal relationships. René Descartes extended the program of separability beyond physics into metaphysics itself, dividing the world into thinking substance and extended substance [7] and thereby establishing the independence of the observer from the observed as a philosophical commitment rather than merely a practical convenience. Isaac Newton's mechanics, published in the *Principia* [8], provided what appeared to be a complete vindication of this program: a deterministic framework in which the state of any system at any time could be computed from its state at any earlier time, given complete knowledge of the forces acting upon it, with no reference to the act of observation or the nature of the observer.

For more than two centuries this framework proved extraordinarily productive, and the successes of classical mechanics, electromagnetism, and thermodynamics reinforced the conviction that separability and determinism were not merely useful approximations but features of the world itself. The first serious challenge came in the early twentieth century with the development of quantum mechanics, but even within quantum theory the implications of inseparability were contested. Heisenberg argued that the act of measurement necessarily disturbs the system being observed, a position that preserves the assumption that properties exist prior to measurement even if we cannot access them without perturbation. Bohr maintained a more radical view, that certain physical quantities are genuinely indeterminate prior to measurement and that the question of what value a property had before it was observed is not merely unanswerable but meaningless. Einstein, Podolsky, and Rosen attempted to demonstrate that quantum mechanics must be incomplete on precisely these grounds, arguing that spatially separated systems must possess definite properties independent of what is measured elsewhere [9]. The concept of separability was central to Einstein's objection, as Howard [10, 11] has shown in detail. Bell's theorem [12, 13] and the experiments that followed [14, 15] resolved the empirical question decisively in favor of Bohr [16]: no theory in which measured quantities have pre-existing, context-

independent values can reproduce the observed correlations. The Kochen-Specker theorem [17] established a related but distinct result, showing that quantum mechanics is contextual even for individual systems, independent of entanglement. Subsequent decades of increasingly stringent experimental tests have continued to confirm Bell violations while closing the loopholes that might have permitted a classical explanation [18–21], and the development of quantum information theory and quantum computing [22–24] has demonstrated that inseparability is not merely an interpretive curiosity but a practical resource, with the Tsirelson bound [25] quantifying the maximum advantage that quantum correlations can provide over classical ones.

The study of language has developed under the same separability assumptions, though the connection is rarely made explicit. The principle of compositionality, which holds that the meaning of a complex expression is determined entirely by the meanings of its parts and the rules by which they combine [26, 27], treats semantic content as a property of linguistic constituents that can be analyzed independently of the interpreter and the context of interpretation. Whether this principle is adequate as a foundation for the study of meaning has been contested on philosophical grounds for more than a century [28–30], but the computational study of language has largely proceeded as if it were settled. Zellig Harris showed that the distributional properties of words in a corpus could serve as a proxy for their semantic relationships [31], and this insight carried through the twentieth century into latent semantic analysis, word embeddings [32, 33], and ultimately large language models [34–36]. At each stage, the underlying computational assumption has been the same: words have meanings that can be represented as fixed points in a real-valued vector space, and the task of a model is to learn where those points are and how they compose. The transformer architecture [37, 38] is the most successful embodiment of this program.

Transformer-based large language models have largely succeeded in passing the well-established benchmarks of artificial intelligence, including conversational assessments that would have been considered definitive demonstrations of language understanding a decade ago [36]. However, their adoption in domains that require guaranteed reliability has been hindered by persistent difficulties, most prominently hallucination [39, 40] and susceptibility to prompt injection [41, 42], which have resisted solution despite substantial engineering effort across architectures and scales. The improvements in capability that once accompanied increases in model size and training data [43, 44] have plateaued [45, 46], and the frontier of the field has shifted toward test-time compute [47], chain-of-thought reasoning, and agentic iteration as strategies for navigating the space of possible responses rather than producing the correct one directly. This shift is consistent with the observation that the informational burden of disambiguating a semantic expression grows superlinearly with its complexity [48], making the recovery of a single intended meaning from an expression of even moderate depth an intractable problem in the sense of relevance realization [49, 50]: the system cannot determine what is relevant from the input alone and must instead explore. Efforts to understand these systems through mechanistic interpretability, which attempts to decompose the internal representations of neural networks into individually meaningful components, have encountered difficulties that appear to be structural rather than merely technical. Sparse autoencoders trained on frontier models lose approximately 90% of the model's capability when their reconstructions replace the original activations [51], extracted features have been shown to be "neither selective nor independent" when used for steering [52], and a recent review with twenty-nine co-authors described the field's foundational concepts as "not yet established" and its status as "pre-paradigmatic" [53]. Theoretical work has demonstrated an exponential gap between the complexity of representing features in superposition and computing with them [54], and sparse autoencoders have been proven to fail to recover ground-truth features except under conditions of extreme sparsity [55]. The difficulty of producing reproducible, meaningful decompositions of neural network representations mirrors the experience of cognitive neuroscience, where decades of functional neuroimaging have shown that localized brain-behavior associations require sample sizes orders of magnitude larger than most studies have used [56], that seventy independent teams analyzing the same fMRI dataset reach substantially different conclusions about which brain regions are involved [57], and that inferring cognitive processes from regional activation is logically unreliable because brain regions participate in many functions simultaneously [58, 59]. In both cases, the assumption that the system can be understood by decomposing it into separable, localizable components has produced results that do not replicate.

In physics, the distinction between a decomposition that has not yet been found and one that cannot exist in principle, because the underlying properties are fundamentally indeterminate prior to measurement, is precisely what Bell's theorem was designed to settle, and the same framework can be applied to semantic interpretation. When the CHSH test is applied to human semantic judgments, the correlations between interpretations produced under different contextual framings violate the classical bound [60–65], and when the same tests are applied to large language models trained on text that human cognition produced, the violations persist across four orders of magnitude in parameter count, with the distributional character of the contextuality orthogonal to every standard benchmark tested [48, 66]. Sheaf-theoretic analysis of BERT's internal representa-

tions has identified over 77,000 instances of contextuality at the level of the embeddings themselves [67, 68]. The non-separability is not confined to the behavioral outputs of these systems; it is present in the geometry of their learned representations [69, 70]. Reconsidered under the premise that meaning is indeterminate prior to the act of interpretation and that natural language is semantically degenerate [48], it necessarily follows that hallucinations and jailbreaks are not anomalies to be eliminated but commonplace consequences of a system that interprets rather than retrieves [66]. If the correlational structure of language is genuinely non-classical, the natural mathematical framework for describing it is the same one that was developed for quantum mechanics: a complex Hilbert space in which states carry phase, similarities are computed through the conjugate inner product, and interference between components is an intrinsic property of the algebra rather than a behavior that must be learned. Large language models built on real-valued representations and softmax attention may functionally replicate this structure, but they do so in the way that any classical simulation of a quantum system does: by using enough parameters to project the complex-valued correlations onto a real-valued space, at a cost in capacity and efficiency that grows with the complexity of the structure being represented.

The representation of signals in complex form has a long history in engineering and physics. Gabor [71] introduced the analytic signal in his theory of communication, and Oppenheim and Lim [72] demonstrated that phase carries more structural information than magnitude in both images and audio [72]. The geometric phase discovered independently by Pancharatnam in optics [73] and Berry in quantum mechanics [74] showed that phase relationships encode topological properties of the space traversed by a system, information that is lost entirely when the representation is projected onto real-valued magnitudes. Complex-valued neural networks have been developed along these lines for decades [75–79], and the holographic reduced representations introduced by Plate [80] demonstrated that complex multiplication and conjugation provide a natural algebra for binding and retrieving associations [81, 82]. Danihelka *et al.* [83] incorporated this algebra into an LSTM with complex-valued cell states, and Ramsauer *et al.* [84] showed that the mathematical structure underlying softmax attention is a modern Hopfield network [85, 86] whose linear variant is the fast weight programmer [87, 88]. None of these efforts, however, has produced a complete language model that operates in complex space from embedding through retrieval to output at a scale where comparison with conventional architectures is meaningful.

Separately, the development of efficient alternatives to the transformer's attention mechanism has produced a body of work that provides the architectural scaffolding for such a model. The transformer [37] computes attention as a softmax-normalized dot product between real-valued projections of the input, an operation that is powerful but quadratic in sequence length and requires a key-value cache that grows linearly during inference. Removing softmax yields a recurrence with matrix state $S_t = S_{t-1} + V_t K_t^\top$ [89], which Schlag *et al.* [88] showed is equivalent to the fast weight programmer introduced by Schmidhuber [87], an associative memory that accumulates associations via outer products and retrieves via matrix-vector product. Subsequent work has refined this structure in various ways: RetNet [90] adds exponential decay, GLA [91] introduces data-dependent gating, DeltaNet [92] replaces additive accumulation with a delta rule, and GateLoop [93] uses complex-valued gates. From the state-space model side, the Linear Recurrent Unit [94] established the importance of complex-valued diagonal recurrences for stable long-range modeling, Mamba [95] introduced input-dependent selection, Griffin [96] validated gated linear recurrence at scale, and Mamba-2 [97] proved the formal equivalence between structured SSMs and linear attention. From the LSTM lineage, mLSTM [98] independently arrives at the same matrix-state recurrence, and RWKV [99, 100] has demonstrated this family at up to 14B parameters. With few exceptions, these models operate in real-valued space. Ramsauer *et al.* [84] showed that softmax attention implements a modern Hopfield network, and the linear variant of this associative memory is precisely the fast weight programmer that the matrix-state models generalize.

To address these shortcomings, we introduce Phase-Associative Memory (PAM), a sequence model that takes the matrix-state structure shared by these lineages and moves it into complex Hilbert space. The state, keys, values, and queries are all complex-valued, and retrieval uses the conjugate inner product $K^* \cdot Q$ rather than the standard dot product, so that the selectivity of retrieval depends on the phase alignment between stored and queried representations. The conjugate inner product is the same operation that computes measurement probabilities in quantum mechanics, and its capacity for constructive and destructive interference is not a learned behavior but an intrinsic property of the algebra.

The architecture emerged through a series of experiments in which each failure was informative. Early versions introduced tokens in complex phase space but destroyed phase information by passing representations through real-valued nonlinearities; correcting this with phase-preserving primitives materially improved results. A subsequent attempt to inject holographic key–value bindings into a vector-state SSM caused a regression in perplexity, because multiple bindings superposed in a single d-dimensional vector interfere destructively with the classical $O(1/\sqrt{n})$ capacity degradation [80]. PAM resolves this by upgrading the state from $\mathbb{C}^d$ to $\mathbb{C}^{d\times d}$, providing $O(d^2)$ associative capacity per head. The reported configuration interleaves channel mixing and PAM in

each of 16 blocks with complex rotary position embeddings [101] on queries and keys, and admits a dual computational form that is $O(T^2)$ for parallel training and $O(1)$ per token for recurrent inference with no KV cache.

At ∼100M parameters on WikiText-103, PAM reaches validation perplexity 30.0 after 10 epochs on a single RTX 4090. A matched transformer trained under identical conditions reaches 27.1. The gap is ∼10%, achieved with a first-generation pure PyTorch implementation, no custom CUDA kernels, and 4× the arithmetic overhead of real-valued computation. The remainder of this paper presents the architecture in detail (), the experimental setup and results (**??**), an analysis of what the progression from vector-state to matrix-state models reveals about the role of associative capacity and phase in sequence modeling (), and a discussion of the limitations of the current work and the questions it leaves open.

METHOD

Overview

The model consists of a complex-valued embedding layer, 16 identical blocks, and a tied complex output head. Each block applies channel mixing via a ComplexGatedUnit (CGU) followed by sequence mixing via a Phase-Associative Memory (PAM) layer, both with residual connections and learned scaling. An earlier configuration stacked all CGU layers before all PAM layers; the interleaved layout reported here, in which every block contains both channel and sequence mixing, produced substantially better results ().

All operations in the main signal path are complex-valued and phase-preserving. Gates and decay parameters use real-valued projections over magnitude features, but the primary data path never converts complex representations to real-valued intermediate forms; doing so would destroy the phase relationships that the conjugate inner product depends on for retrieval.

Phase-preserving primitives

Complex representation. Tensors representing complex quantities have shape $[\ldots, d, 2]$, implementing $\mathbb{C}^d$ in split-real form.

Complex linear map. Given weight matrices $W_r, W_i \in \mathbb{R}^{m\times n}$, the complex linear map computes:

$$y_r = W_r x_r - W_i x_i, \quad y_i = W_i x_r + W_r x_i. \tag{1}$$

modReLU. A phase-preserving activation that thresholds magnitude while leaving phase untouched:

$$\mathrm{modReLU}(z) = \mathrm{ReLU}(|z| + b) \cdot \frac{z}{|z|} \tag{2}$$

where b is a learned bias.

ComplexNorm. RMS normalization applied to magnitudes, with phase preserved:

$$\mathrm{ComplexNorm}(z) = s \cdot \frac{|z|}{\mathrm{RMS}(|z|)} \cdot \frac{z}{|z|} \tag{3}$$

where s is a learned scale parameter.

ComplexGatedUnit (CGU). A SwiGLU-style gating block in complex space. The gate magnitude $\sigma(|W_g z|)$ controls how much signal passes, while the gate phase controls what rotation is applied. An up-projection through modReLU provides the nonlinearity:

$$\mathrm{CGU}(z) = W_{\mathrm{down}}\big(\mathrm{gate}_{\mathrm{phase}} \odot \mathrm{modReLU}(W_{\mathrm{up}} z) \cdot \sigma(|W_g z|)\big) \tag{4}$$

Blocks (interleaved CGU and PAM)

Each of 16 blocks applies **channel mixing** then **sequence mixing** with residual connections and learned scaling:

$$\tilde{z}^{(l)} = z^{(l-1)} + \alpha^{(l)}_{\mathrm{CGU}} \cdot \mathrm{CGU}_l(\mathrm{ComplexNorm}(z^{(l-1)})), \tag{5}$$

$$z^{(l)} = \tilde{z}^{(l)} + \alpha^{(l)}_{\mathrm{PAM}} \cdot \mathrm{PAM}_l(\mathrm{ComplexNorm}(\tilde{z}^{(l)})) \tag{6}$$

where $\alpha^{(l)}_{\mathrm{CGU}}$ is initialized to 1.0 and $\alpha^{(l)}_{\mathrm{PAM}}$ to 0.1. Each CGU uses expansion factor 3 (hidden dimension $3\times384 = 1152$). Dropout is applied during training.

Phase-Associative Memory (PAM)

PAM replaces both the recurrent backbone and the attention mechanism with a single module whose operations correspond directly to the quantum semantic framework described in [48]. In that framework, a semantic expression S_E is represented as a state vector $|\psi_{S_E}\rangle = \sum_i c_i |e_i\rangle$ in a complex Hilbert space, where the complex coefficients c_i carry phase information with no classical analogue, and interpretation is the application of a Hermitian operator whose eigenstates represent possible meanings. PAM implements this structure computationally: tokens are embedded as complex vectors, associations between them are accumulated in a complex matrix state via outer products, and retrieval is the projection of a query onto the accumulated state through the conjugate inner product, the same operation that computes $P(m_i) = |\langle e_i|\psi_{S_E}\rangle|^2$ in the quantum semantic framework.

Each PAM head h maintains a complex matrix state $S_t^{(h)} \in \mathbb{C}^{d\times d}$, where d is the head dimension, so that the total state capacity across H heads is $H \times d^2$ complex values per layer ($6 \times 64^2 = 24{,}576$ in our configuration).

The input $x_t \in \mathbb{C}^D$ is projected into queries, keys, and values via a single complex linear map:

$$[Q_t; K_t; V_t] = W_{\text{QKV}} x_t \quad \Rightarrow \quad Q_t, K_t, V_t \in \mathbb{C}^{H \times d}. \quad (7)$$

Complex rotary position embeddings [101] are applied to Q and K by multiplying each element by a precomputed unit-magnitude factor $e^{im\theta}$, encoding absolute position in phase while leaving magnitudes unchanged; in the conjugate product $K_i^* \cdot \tilde{Q}_t$ the dependence on position difference $(m-n)$ yields relative position structure. Retrieval uses the scaled query $\tilde{Q}_t := Q_t/\sqrt{d}$.

The decay rate γ_t controls how quickly the state forgets and is computed from the input as $\gamma_t = \exp(-\operatorname{softplus}(W_{dt} \cdot \operatorname{concat}(x_{t,r}, x_{t,i}) + b_{dt}))$, where b_{dt} is initialized to -4.0 for slow initial decay. A learned protect gate $p_t = \sigma(W_p \cdot |x_t| + b_p)$ with $b_p = -3.0$ modifies the effective decay:

$$\gamma_t = e^{-dt_t} \cdot (1 - p_t) + p_t, \qquad V_t' = V_t \cdot (1 - p_t). \quad (8)$$

When $p_t \to 1$ the state is frozen and new values are suppressed; when $p_t \to 0$ the decay proceeds normally. The state then evolves as:

$$S_t = \gamma_t \cdot S_{t-1} + V_t' \otimes K_t^* \quad (9)$$

where $\otimes$ denotes complex outer product and K_t^* is the complex conjugate of the key. Retrieval computes $Y_t = S_t \tilde{Q}_t$, which expands to:

$$Y_t = \sum_{i \le t} \left(\prod_{j=i+1}^{t} \gamma_j \right) \left(K_i^* \cdot \tilde{Q}_t \right) V_i'. \quad (10)$$

The conjugate inner product $K_i^* \cdot \tilde{Q}_t$ determines retrieval strength through phase alignment: associations whose keys are phase-coherent with the query are retrieved strongly while phase-incoherent associations are suppressed, without softmax normalization.

Dual form (training). During training, the recurrence is computed in $O(T^2)$ time by forming the decay matrix $D \in \mathbb{R}^{T \times T}$ with $\log D[t,i] = \sum_{j=i+1}^{t} \log \gamma_j$ via cumulative sums, applying a causal mask, computing the complex score matrix $W = \tilde{Q} K^{*\top}$, and obtaining the output as $Y = (W \odot D) \cdot V'$. This is mathematically equivalent to the recurrence but parallelizes across the sequence dimension. The final state $S_T = \sum_{i=1}^{T} D[T,i] \cdot (V_i' \otimes K_i^*)$ is computed at the end of each forward pass so that generation can continue from a prompt.

Recurrent form (inference). During autoregressive generation, each token requires $O(Hd^2)$ work per layer: compute Q_t, K_t, V_t', γ_t; apply RoPE; update $S_t = \gamma_t S_{t-1} + V_t' \otimes K_t^*$; retrieve $Y_t = S_t \tilde{Q}_t$. The state $S \in \mathbb{C}^{H \times d \times d}$ is fixed-size and does not grow with sequence length.

TABLE I. medium-pam-v3 configuration (reported).

Parameter	Value
Complex dimension (D)	384
Blocks	16 (each: CGU → PAM)
Interleaved CGU + PAM	yes
CGU expansion factor	3
PAM heads (H)	6
PAM head dimension (d)	64
Gated State Protection	enabled
Complex RoPE on Q, K	yes
QK phase normalization	off (see)
Sequence length	2048
Total parameters	∼100.4M

Output head

Logits are computed via a tied complex inner product with the embedding table: $\text{logits} = \operatorname{Re}(z_{\text{out}} \cdot \operatorname{conj}(E)) = z_{\text{out},r} \cdot E_r^\top + z_{\text{out},i} \cdot E_i^\top$, where E is shared with the input embedding.

EXPERIMENTAL SETUP

Dataset

We train and evaluate on WikiText-103 [102], a standard language modeling benchmark containing approximately 103 million tokens of Wikipedia text. We use the official train/validation splits, tokenized with the GPT-2 BPE tokenizer (vocabulary size 50,257).

Model configuration

Primary reported model (medium-pam-v3). See I.

Training details

Training hyperparameters are listed in II. The earlier sequential configuration used a lower learning rate (3×10^{-5}) and shorter warmup (500 steps); see .

Generation settings (qualitative evaluation)

During training, generation samples are logged every 5,000 steps using: temperature 1.0; top-k 50; top-p 0.9; repetition penalty 1.2.

TABLE II. Training hyperparameters (medium-pam-v3).

Parameter	Value
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	0.01
Warmup steps	1000
LR schedule	warmup + cosine decay
Batch size	3
Epochs	10
Gradient clipping	1.0
Precision	automatic mixed precision (bf16)
Hardware	single NVIDIA RTX 4090 (24GB)
Compilation	torch.compile (default mode)
Initialization	orthogonal (complex linear maps)

TABLE III. medium-pam-v3: training and validation perplexity by epoch (WikiText-103).

Epoch	Train PPL	Val PPL
1	123.86	57.94
2	53.87	43.83
3	44.88	38.69
4	40.39	35.88
5	37.42	33.82
6	35.13	32.25
7	33.26	31.22
8	31.78	30.40
9	30.66	30.01
10	30.02	**30.0**

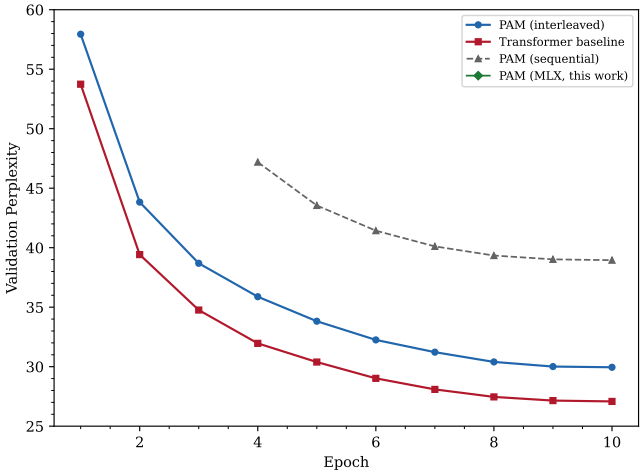

FIG. 1. Validation perplexity vs. epoch for the interleaved PAM configuration, the sequential PAM configuration, and the matched transformer baseline, all trained on WikiText-103.

TABLE IV. Matched comparison: transformer baseline vs. PAM (same data, tokenizer, seq_len, LR, hardware).

Model	Type	Params	Val PPL	Wall time
GPT-2 (cited)	Transformer	124M	∼31	–
Transformer (ours)	Transformer	∼100.3M	**27.1**	3.5 h
PAM (ours)	**Complex recurrent**	**∼100.4M**	**30.0**	**14.1 h**

RESULTS

Main result

The interleaved configuration with complex RoPE reaches validation perplexity **30.0** after 10 epochs on WikiText-103 (single run, RTX 4090). Fig. 1 and Table III summarize training progress.

Total wall time **50,714 s (∼14.1 h)**. Throughput **∼23k tokens/second** average.

QK phase normalization ablation. We briefly trained with per-element unit normalization of Q and K before the conjugate inner product. Validation loss decreased, but generation collapsed into severe lexical repetition by mid-training and the run was stopped during epoch 5. Because PAM scores are not softmax-normalized, removing the magnitude degree of freedom from Q and K eliminates the model's ability to modulate retrieval strength, leaving only phase alignment to determine selectivity. The reported configuration does not use QK phase normalization.

Ablations. An earlier sequential configuration (16 CGU layers followed by 16 PAM layers, no RoPE, lower learning rate) achieved 38.95 validation PPL, and a hybrid that adds sparse windowed attention every 4th block achieved 30.01, marginally worse than pure PAM. These results are not controlled single-variable ablations, but they indicate that interleaving channel and sequence mixing is important and that supplemental attention provides no benefit at this scale.

Comparison to matched transformer baseline

To provide a rigorous comparison, we trained a standard transformer with ∼100.3M parameters on the same WikiText-103 data, tokenizer (GPT-2 BPE), sequence length (2048), batch size (3), learning rate (1×10^{-4}), warmup (1000 steps), and hardware (single RTX 4090). The transformer uses $d_{\text{model}} = 672$, 12 layers, 12 attention heads, and $d_{\text{ff}} = 2688$.

The ∼10% gap in perplexity is accompanied by a ∼4× gap in training throughput (23k vs. 96k tok/s), which is expected given that complex linear maps require four real matrix multiplies and the PAM implementation uses no custom CUDA kernels. For external reference, GPT-2 [103] at 124M parameters achieves ∼31 validation PPL on WikiText-103, though differences in evaluation protocol make direct comparison unreliable.

The transformer converges faster and maintains a consistent advantage throughout training (Table V), though

TABLE V. Epoch-by-epoch validation PPL: matched transformer vs. medium-pam-v3.

	Transformer (ours)		PAM (ours)	
Epoch	Train PPL	Val PPL	Train PPL	Val PPL
1	137.77	53.74	123.86	57.94
2	52.81	39.42	53.87	43.83
3	42.74	34.76	44.88	38.69
4	38.27	31.96	40.39	35.88
5	35.54	30.39	37.42	33.82
6	33.41	29.02	35.13	32.25
7	31.65	28.09	33.26	31.22
8	30.28	27.46	31.78	30.40
9	29.29	27.15	30.66	30.01
10	28.75	**27.1**	30.02	**30.0**

the gap narrows in later epochs. The two models see the same total tokens. Despite the perplexity gap, the structural differences between the architectures are substantial: the transformer requires $O(T)$ work per token during inference because of the KV cache, whereas PAM's state is fixed at 49,152 floats per layer regardless of sequence length, giving it $O(1)$ per-token cost with per-layer work $O(Hd^2)$. At $T = 2048$ with the baseline's configuration, the transformer's KV cache per layer is $\sim$56$\times$ larger than PAM's state, and this ratio grows linearly with context length.

Generation quality

medium-pam-v3 (epoch 10) produces coherent multi-sentence text; factual accuracy is not guaranteed.

Prompt: `In 1923 , the University of`

Generated text (medium-pam-v3, epoch 10):

> In 1923 , the University of Illinois at Urbana @-@ Urdu said it was " an easy choice to do something in its own right . " The university also claimed the first students from Wisconsin had to be replaced by a more " good student " due to a lack of funds .

Quality metrics at final epoch: 3-gram repetition rate **0.034**, 4-gram repetition rate **0.011**, zero restarts, unique token ratio **0.703**. The sample shows section-like structure and WikiText-style tokenization artifacts; it does **not** exhibit the list-repetition failure mode of the QK-norm ablation.

Sequential medium-pam (epoch 10) for the same prompt (different run, see):

> In 1923 , the University of Missouri and the University of Michigan was also established in 1926 . In 1928 , the University of Michigan opened its current campus with the school 's first campus opening at St. Louis Road on the northern end of Lake Michigan in 1929 .

Quality metrics (sequential run): 3-gram rep. 0.051, 4-gram rep. 0.020, unique token ratio 0.624.

The text is grammatically coherent and shows structural awareness (dates, proper nouns, institutional language). Factual accuracy is not reliable—this is expected at this scale.

Earlier validation: TinyStories

Before WikiText-103, the broader V6 architecture family was validated on TinyStories [104]:

Model	Config	Params	Dataset	V
V6 small-matched	CGU + SSM, no memory	28.7M	TinyStories (full)	
V6 small-matched	CGU + SSM, WM=16	29.2M	TinyStories (full)	

These results established that the phase-preserving, attention-free architecture family can learn effectively. The TinyStories experiments also revealed that working memory capacity is a critical control variable: too much memory induces memorization, while zero memory produces cleaner but higher-perplexity generations.

DISCUSSION

Schlag *et al.* [88] showed that linear attention without softmax is equivalent to a fast weight programmer [87] in which outer products of keys and values are accumulated into a matrix that queries retrieve from via matrix-vector product. Subsequent work has refined this structure with exponential decay [90], data-dependent gating [91], delta-rule updates [92], and complex-valued gates [93], while from the LSTM lineage Beck *et al.* [98] arrived at the same recurrence independently. PAM belongs to this family, but the substitution of the conjugate inner product for the standard dot product introduces a qualitative difference that is worth examining in the context of what is known about the limitations of linear attention.

Arora *et al.* [105] demonstrated that linear attention models struggle with associative recall, the ability to retrieve a specific stored value given a matching key, because the non-negative nature of real-valued inner products means that all stored associations contribute positively to the retrieval, diluting the target signal. Ramsauer *et al.* [84] showed that softmax attention avoids this problem because it implements a modern Hopfield network [85, 86] in which the exponential nonlinearity sharpens retrieval to a near-one-hot selection. PAM's conjugate inner product provides a different mechanism for addressing the same problem: because the complex dot product $K_i^* \cdot \tilde{Q}_t$ can be negative or imaginary depending on the phase relationship between the stored key and the query, associations that are phase-incoherent

with the query are actively suppressed rather than merely downweighted. Whether this destructive interference is as effective as the exponential sharpening of softmax at recovering specific associations from a large store is a question that passkey retrieval experiments at long context lengths would address directly.

Danihelka *et al.* [83] previously incorporated complex-valued holographic representations into an LSTM and found improvements on tasks requiring compositional binding and retrieval, consistent with the classical result of Oppenheim and Lim [72] that phase carries more structural information than magnitude. The connection between phase and representational capacity has a deeper grounding in the geometric phase of Berry [74] and Pancharatnam [73], which showed that phase relationships encode topological properties of the trajectory a system follows through its state space. In the context of PAM, this suggests that the phase of the complex state may encode not only the content of stored associations but something about the path through which they were accumulated, a form of sequential information that real-valued states cannot represent without additional mechanisms. The QK phase normalization ablation described in , in which removing the magnitude degree of freedom from queries and keys caused generation to collapse into repetition, provides indirect evidence that both magnitude and phase are carrying distinct and necessary information in the retrieval process.

Lo *et al.* [67] used a sheaf-theoretic framework [68] to identify over 77,000 instances of contextuality in BERT's internal representations, demonstrating that the non-separability observed behaviorally in Bell violation experiments [48, 60, 64, 66] is also present in the geometry of the embeddings themselves. This finding connects to the difficulties documented in the mechanistic interpretability literature, where sparse autoencoders trained on frontier models lose approximately 90% of the model's capability when their reconstructions replace the original activations [51], extracted features are neither selective nor independent when used for steering [52], and there is a proven exponential gap between representing features in superposition and computing with them [54]. Williams *et al.* [69] have argued that these difficulties may reflect conceptual limitations of the decomposition framework rather than technical ones. If the representations are fundamentally non-separable, as the Bell violations and the sheaf-theoretic analysis suggest, then an architecture that operates natively in complex Hilbert space may not merely provide an alternative computational mechanism but may produce representations whose structure is more naturally described by the formalism in which they were computed. Whether PAM's representations are in fact more interpretable in sheaf-theoretic terms than transformer representations is a testable question that we intend to pursue.

The computational costs are comparable in order to standard attention: the training-time dual form gives $O(T^2Hd)$ per layer, while at inference PAM requires $O(Hd^2)$ per token using a fixed state of 49,152 floats per layer regardless of sequence length, compared to a KV cache that grows linearly with context. All results reported here are from single training runs on a single RTX 4090, and the 4× throughput gap relative to the transformer (23k vs. 96k tokens/second) reflects the arithmetic cost of complex linear maps and the absence of custom CUDA kernels rather than a fundamental asymmetry in computational complexity.

CONCLUSION

In this work we have constructed a language model whose representations, projections, activations, and retrieval all operate in complex-valued Hilbert space, and evaluated it against a matched transformer on a standard language modeling benchmark. The architecture, Phase-Associative Memory, accumulates associations in a complex matrix state $S_t \in \mathbb{C}^{d\times d}$ via outer products and retrieves via the conjugate inner product $K^* \cdot Q/\sqrt{d}$, the same operation that computes projection probabilities in the quantum semantic framework [48]. At ∼100M parameters on WikiText-103, PAM reaches validation perplexity 30.0, within ∼10% of a matched transformer (27.1) trained under identical conditions, despite 4× arithmetic overhead and no custom kernels.

The development of the architecture revealed that holographic binding in vector-state models fails due to the $O(1/\sqrt{n})$ capacity degradation of superposed associations, and that upgrading to a matrix state resolves this; that magnitude and phase must both be free to vary for the conjugate inner product to function as a retrieval mechanism without softmax; and that supplementing PAM with sparse attention provides no benefit at this scale. These findings are consistent with the hypothesis that the complex recurrence is already capturing the correlational structure that attention exploits, though the ∼10% gap indicates that the capture is not yet complete.

Whether the gap reflects an implementation limitation, a fundamental advantage of attention over recurrence, or the difference between native complex-valued computation and its real-valued approximation is a question that the complex-vs-real ablation at matched parameter count is designed to address. The scaling behavior of the gap at 350M and 1B parameters would determine whether the complex-valued formalism captures a progressively larger share of the structure of language as models grow. The CHSH contextuality evaluation developed in [66] has been applied to transformer-based models but not yet to PAM, and running the semantic Bell test on PAM-generated interpretations would test whether an architecture built on the quantum semantic

formalism produces a contextuality profile different from one that must approximate the same structure through real-valued attention.

Additional Details

The matched transformer baseline uses $d_{\text{model}} = 672$, 12 layers, 12 attention heads, and $d_{\text{ff}} = 2688$ ($\sim$100.3M parameters). The earlier sequential PAM configuration uses the same embedding and head dimensions as the reported interleaved model but without interleaving, without RoPE, and with learning rate 3×10^{-5} and 500 warmup steps.

The embedding accounts for a large fraction of PAM's parameters because dim = 384 with vocabulary size 50,257 requires $50{,}257 \times 384 \times 2 = 38.6$M parameters for the real and imaginary embedding tables, which are tied with the output layer. The remaining 61.7M parameters are in the 16 CGU + PAM blocks.

Reproducibility

Full source code and training scripts: `https://github.com/gowrav-vishwakarma/qllm2`.

Python 3.9+ (tested with 3.9.24); PyTorch 2.x with CUDA support; package management via `uv`.

The reported **30.0** validation PPL run is archived under `logs/v6/medium_pam_v3_rope_wikitext103_20260319_231524_31397f0/` (`RUN_INFO.txt` lists git commit `31397f0` and full CLI). The matched transformer baseline (**27.1**) is archived under `logs/v6/transformer_baseline_wikitext103_20260323_140351_e87b8e4/` (git commit `e87b8e4`).

```
uv sync && uv sync --extra cuda
./scripts/run_v6_medium_pam_v3.sh
./scripts/run_v6_medium_pam.sh
./scripts/run_v6_medium_pam.sh --resume
```

Generation:

```
python -m v6.generate \
  --checkpoint checkpoints_v6_medium_pam_v3/best_model.pt \
  --prompt "In 1923 , the University of"
```


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